



CSE-476: Introduction to Natural Language Processing
Quiz 2

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. What is the main difference between generative and discriminative classifiers?
- (A) Generative models model $P(x, y)$, while discriminative models model $P(y | x)$
 - (B) Discriminative models learn the data distribution and use that knowledge for classification
 - (C) Discriminative models require no training data
 - (D) Generative models are less explainable
- Q2. Which technique is used to mitigate overfitting by penalizing large weights?
- (A) Laplace smoothing
 - (B) Regularization
 - (C) TF-IDF
 - (D) Softmax normalization
- Q3. What is the limitation of the perceptron algorithm?
- (A) It cannot use vector representations
 - (B) It only works for linearly separable data
 - (C) It requires probabilistic outputs
 - (D) It cannot be trained with gradient descent
- Q4. Why is Laplace smoothing important?
- (A) It improves cosine similarity
 - (B) It reduces dimensionality
 - (C) It avoids zero probabilities for unseen words
 - (D) It speeds up gradient descent
- Q5. What happens in stochastic gradient descent (SGD) compared to gradient descent (GD)?

- (A) SGD updates weights using the entire dataset every step
- (B) SGD updates weights using one or a subset of training examples
- (C) GD converges faster because updates are noisier
- (D) SGD cannot be used with neural networks

Q6. What is the core intuition behind word embeddings?

- (A) Words are represented by their dictionary definitions
- (B) Words that appear in similar contexts have similar meanings
- (C) Words are clustered by part of speech
- (D) Words are ranked by frequency

Q7. What is the primary purpose of the Softmax function?

- (A) To penalize large model weights
- (B) To reduce feature dimensionality
- (C) To select the class with the largest raw score without normalization
- (D) To convert raw class scores into a valid probability distribution

Q8. In the perceptron algorithm, when are the weights updated?

- (A) After every training example
- (B) Only when the prediction is correct
- (C) Only when the prediction is incorrect
- (D) Only after one full pass over the data

Q9. What is the main goal of TF-IDF weighting?

- (A) Increase the weight of very common words
- (B) Represent word order
- (C) Convert sparse vectors into dense embeddings
- (D) Highlight words that are frequent in a document but rare across documents

Q10. Why is cosine similarity preferred over dot product when comparing word vectors?

- (A) It ignores vector length while calculating similarity
- (B) It increases the influence of high-frequency words
- (C) It converts similarity scores into probabilities
- (D) It reduces the dimensionality of word embeddings

End of Paper